



Cloud learning and global intelligence

seconds–hours



Foundation-model
training / updates



Fleet memory



Simulation &
digital twins



Global task planning



Model/skill updates • compressed context • audit traces



Edge cognition and multi-agent coordination

milliseconds–seconds



Multimodal fusion



VLA / world-model
inference



Model / expert
sharing



Task / physical
interaction graph

Deterministic radio scheduling



Synchronized features • state/action tokens • deadlines • uncertainty



On-device reflex, control, and safety

sub-ms–milliseconds



Sensor acquisition



State estimation



Servo control



Collision avoidance

Fail-safe fallback

Cross-layer orchestrator



Token value



Radio state



Compute queues



Control phase